

IQC:

Block encoding.

$$A \in \mathbb{C}^{N \times N}, \quad N = 2^n$$

how to access A when A is not unitary

If A is Hermitian $A = A^\dagger$

$$U = e^{itA}$$

$$A = V D V^\dagger \quad V V^\dagger = I$$

$$\triangleq V e^{itD} V^\dagger$$

$$\hookrightarrow \begin{bmatrix} e^{it\lambda_1} & & 0 \\ & e^{it\lambda_2} & \\ 0 & & \ddots \\ & & & e^{it\lambda_n} \end{bmatrix}$$

If A is not Hermitian (doubling reasoning)

$$\tilde{A} = \begin{bmatrix} 0 & A^\dagger \\ A & 0 \end{bmatrix}$$

$$\tilde{A}^\dagger = \tilde{A} \quad \text{is Hermitian}$$

We can design a circuit from e^{itA} using Trotterisation

Block encoding $\leftarrow$ If A is dense or not data-sparse any quantum input model would need an exponential number of operations to implement it.

$$A = X Y^T = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} y_1 & \dots & y_n \end{bmatrix} \quad \text{is data sparse}$$

the usual assumption is that A is s -sparse or
 data sparse $\downarrow \text{nnz}(A) / \# \text{rows} \approx s$

Query model: assume to have an oracle such that

Assume $\|A\|_{\max} = \max_{i,j} |A_{ij}| \leq 1$ $\tilde{A}/d$ $d \geq \|A\|_{\max}$

$$O_A |0\rangle |i\rangle |j\rangle = \underbrace{(A_{ij} |0\rangle + \sqrt{1 - |A_{ij}|^2} |1\rangle)}_{\text{signal qubits}} \underbrace{|i\rangle |j\rangle}_{\text{data-qubits}}$$

This oracle can be decomposed into two parts

① $\tilde{O}_A |0\rangle^d |i\rangle |j\rangle = |\tilde{A}_{ij}\rangle |i\rangle |j\rangle$
 $|\tilde{A}_{ij}| - |A_{ij}| < 2^{-d}$ — binary rep of A_{ij} over d -digits
 + amplitude estimation (QFT)

② otherwise using some classical arithmetic operators

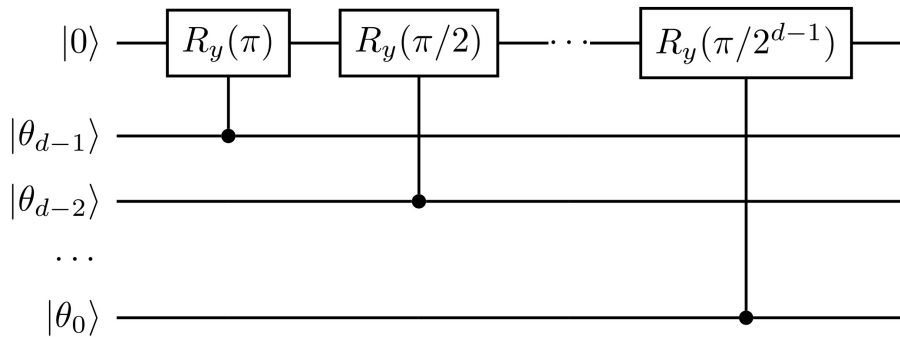
$$O'_A |0\rangle^{d'} |i\rangle |j\rangle = |\tilde{\vartheta}_{ij}\rangle |i\rangle |j\rangle$$

$\tilde{\vartheta}_{ij}$ is a d' -bit representation of $\vartheta_{ij} = \frac{1}{\pi} \arccos A_{ij}$

Now with controlled rotations we can put the $\tilde{A}_{ij}$ inside the amplitude of an ancillary qubit.

$$U_{ce} |0\rangle |\tilde{\psi}_{ij}\rangle = (A_{ij} |0\rangle + \sqrt{1 - |A_{ij}|^2} |1\rangle) |\tilde{\psi}_{ij}\rangle$$

The quantum circuit for the controlled rotation circuit is



$$|0\rangle |0\rangle^{d'} |i\rangle |j\rangle \xrightarrow{O_A^1} |0\rangle |\tilde{\psi}_{ij}\rangle |i\rangle |j\rangle \xrightarrow{U_{ce}} (A_{ij} |0\rangle + \sqrt{1 - |A_{ij}|^2} |1\rangle) |\tilde{\psi}_{ij}\rangle |i\rangle |j\rangle \xrightarrow{O_A^{1\dagger}} (A_{ij} |0\rangle + \sqrt{1 - |A_{ij}|^2} |1\rangle) |0\rangle^{d'} |i\rangle |j\rangle$$

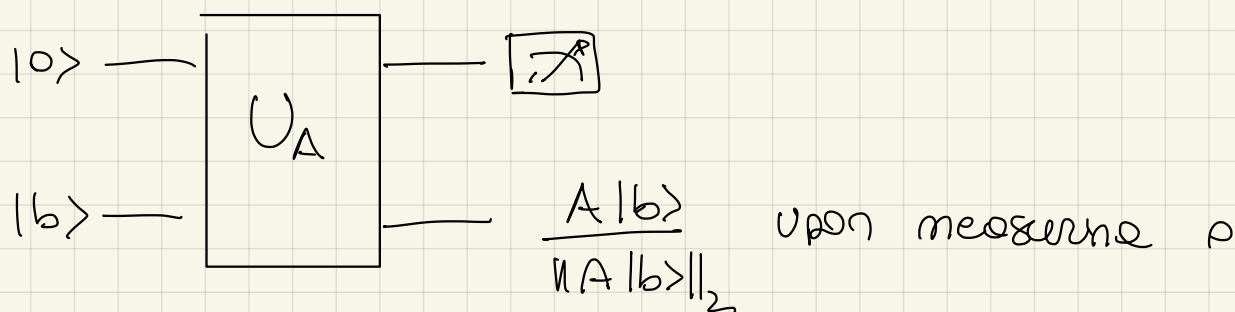
$$U_A = \begin{bmatrix} A & * \\ * & * \end{bmatrix} \quad \begin{array}{l} \text{irrelevant entries} \\ \text{just to complete } A \text{ to} \\ \text{unitary matrix} \end{array}$$

(n+1) - qubit

Ab

$$U_A |0\rangle |b\rangle = \begin{bmatrix} A & * \\ * & * \end{bmatrix} \begin{bmatrix} b \\ 0 \end{bmatrix} = \begin{bmatrix} Ab \\ * \end{bmatrix} = |0\rangle A|b\rangle + |1\rangle |*\rangle$$

To obtain $A|b\rangle$ we need to measure first qubit to zero



$$\text{prob}(0) = \|A|b\rangle\|^2 = \langle b|A^\dagger A|b\rangle$$

the normalizing term $\|A|b\rangle\|_2$ can be estimated from the success probability $\text{prob}(0)$.

We can work with even larger matrices to improve the implementation of U_A (lower the depth of the circuit for U_A)

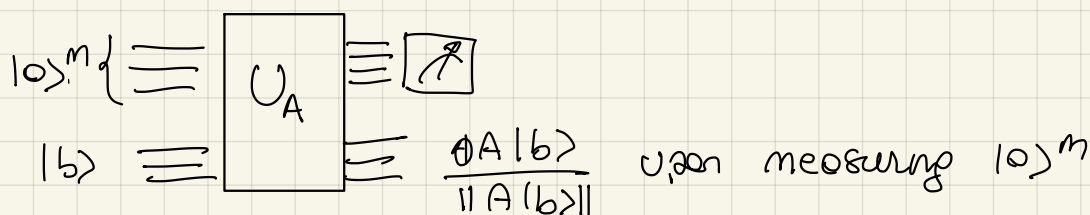
$$U_A = \begin{pmatrix} A & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{pmatrix}$$

using m additional qubits

$$A \in \mathbb{C}^{2^n \times 2^n}$$

$$U_A \in \mathbb{C}^{2^{n+m} \times 2^{n+m}}$$

$$A = \left(\langle 0|^m \otimes I_n \right) U_A \left(|0\rangle^m \otimes I_n \right)$$



$$\tilde{A} = A/\alpha$$

$$\text{prob}(0^m) = \|\tilde{A}|b\rangle\|^2 = \frac{1}{\alpha^2} \|A|b\rangle\|^2$$

If α is too large $p(0^m)$ can be very -very small

Def: Given an n -qubit matrix A ($A \in \mathbb{C}^{N \times N}$, $N=2^n$)
if we find $\alpha, \epsilon \in \mathbb{R}_+$ and an $(n+m)$ -qubit unitary U_A such that

$$\|A - \alpha(\langle 0^m| \otimes I_n) U_A |0^m\rangle \otimes I_n\| \leq \epsilon$$

then U_A is an (α, m, ϵ) -BE(A)

$$U_A = \begin{bmatrix} \hat{A}/\alpha & * & * \\ 1 & & * \\ \vdots & \vdots & \vdots \end{bmatrix} \quad \|A - \alpha \cdot \hat{A}/\alpha\| \leq \epsilon$$

With BE we can access to the entries of A

$$\hat{A}_{ij} = \langle 0^m | i | \underbrace{U_A | 0^m | j \rangle}_{\alpha}$$

$$\begin{array}{c} |0^m\rangle \\ |j\rangle \end{array} \Rightarrow \boxed{U_A} \Rightarrow \boxed{\times} \quad \sum_{i=0}^{N-1} A_{ij} |i\rangle = \begin{bmatrix} A_{0j} \\ A_{1j} \\ \vdots \\ A_{N-1,j} \end{bmatrix}$$

Remark: It's always possible to have a $(\alpha, 1)$ -BE(A)

$$A = W \Sigma V^+ \quad \text{is SVD of } A$$

$$U_A = \begin{bmatrix} W & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \Sigma & \sqrt{I - \Sigma^2} \\ \sqrt{I - \Sigma^2} & -\Sigma \end{bmatrix} \begin{bmatrix} V^+ & 0 \\ 0 & I \end{bmatrix}$$

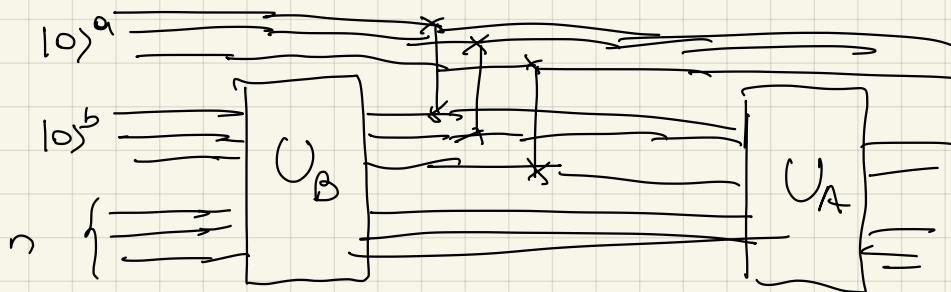
$$= \begin{bmatrix} A & W \sqrt{I - \Sigma^2} \\ \sqrt{I - \Sigma^2} V^+ & -\Sigma \end{bmatrix}$$

This block encoding can be extremely difficult to decompose in terms of elementary quantum gates

$$\text{If } U_A \text{ is } (\alpha, a, E_A) - \text{BE}(A) \quad \begin{bmatrix} A \\ \vdots \end{bmatrix}$$

$$U_B \text{ is } (\beta, b, E_B) - \text{BE}(B) \quad \begin{bmatrix} B \\ \vdots \end{bmatrix}$$

we can construct a $(\alpha\beta, a+b, \alpha E_B + \beta E_A) - \text{BE}(AB)$



$$(I \otimes U_A) (SWAP(a,b) \otimes I_N) (I \otimes U_B)$$

ex: $a=b=1$

$$SWAP = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\left[\begin{array}{c|c} \begin{array}{c|c} A & x \\ \hline x & \delta \end{array} & 0 \\ \hline 0 & \begin{array}{c|c} A & \delta \\ \hline \delta & \delta \end{array} \end{array} \right] \begin{bmatrix} I & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \\ 0 & 0 & 0 & I \end{bmatrix} \left[\begin{array}{c|c} \begin{array}{c|c} B & \delta \\ \hline \delta & \delta \end{array} & 0 \\ \hline 0 & \begin{array}{c|c} B & \delta \\ \hline \delta & \delta \end{array} \end{array} \right] = \left[\begin{array}{c|c} \begin{array}{c|c} AB & x \\ \hline x & \delta \end{array} & \begin{array}{c|c} x & \delta \end{array} \\ \hline \begin{array}{c|c} \delta & \delta \end{array} & \begin{array}{c|c} \delta & \delta \end{array} \end{array} \right]$$

We can design q. circuits for the block encoding of a linear combination of matrices

$$A = \sum_{i=0}^S \gamma_i A_i$$

↑

U_i is (d_i, a_i, ϵ_i) -BE(A_i)

we can construct the BE of A

$$\begin{bmatrix} I \\ X \end{bmatrix}$$